chapter-3

Bayesian Concept Learning

3.1 Importance of Bayesian Methods:

- Bayesian methods allow us to estimate model parameters, to construct model forecasts and to conduct model comparisons. Bayesian learning algorithms can calculate explicit probabilities for hypotheses.
- Bayesian classifiers use a simple idea that the training data are utilized to calculate an observed probability of each class based on feature values.
- When Bayesian classifier is used for unclassified data, it uses the observed probabilities to predict the most likely class for the new features.
- Each observed training example can incrementally decrease or increase the estimated probability that a hypothesis is correct.
- Prior knowledge can be combined with observed data to determine the final probability of a hypothesis. In Bayesian learning, prior knowledge is provided by asserting a prior probability for each candidate hypothesis, and a probability distribution over observed data for each possible hypothesis.
- Bayesian methods can accommodate hypotheses that make probabilistic predictions. New instances can be classified by combining the predictions of multiple hypotheses, weighted by their probabilities.
- Even in cases where Bayesian methods prove computationally intractable, they can provide a standard of optimal decision making against which other practical methods can be measured.
- **Uses of Bayesian classifiers are as follows:**

(1) Used in text-based classification for finding spam or junk mail filtering.

(2) Medical diagnosis.

(3) Network security such as detecting illegal intrusion.

3.2 Bayes Theorem:

- Bayes' theorem is a method to revise the probability of an event given additional information. Bayes's theorem calculates a conditional probability called a posterior or revised probability.
- Bayes' theorem is a result in probability theory that relates conditional probabilities. If A and B denote two events, P(A|B) denotes the conditional probability of A occurring,

given that B occurs. The two conditional probabilities P(A|B) and P(B|A) are in general different.

- Bayes theorem gives a relation between P(A|B) and P(B|A). An important application of Bayes' theorem is that it gives a rule how to update or revise the strengths of evidence-based beliefs in light of new evidence a posteriori.
- A **prior probability** is an initial probability value originally obtained before any additional information is obtained.
- A **posterior probability** is a probability value that has been revised by using additional information that is later obtained.
- Suppose that B1, B2, B3 ... Bn partition the outcomes of an experiment and that A is another event. For any number, k, with $1 \leq k \leq n$, we have the formula:

$$P(B_k/A) = \frac{P(A/B_k) \cdot P(B_k)}{\sum_{i=1}^{n} P(A/B_i) \cdot P(B_i)}$$

Example 3.2.1:

A mechanical factory production line is manufacturing bolts using three machines, A, B and C. The total output, machine A is responsible for 25 %, machine B for 35% and machine C for the rest. The machines that 5 % of the output from machine A is defective, 4% from machine B and 2 % from machine C. A bolt is chosen at random from the production line and found to be defective. What is the probability that it came from

i. machine A ii.machine B iii.machine C ?

Solution : Let

D = {bolt is defective},

A = {bolt is from machine A},

B = {bolt is from machine B},

C = {bolt is from machine C}.

Given data : $P(A) = 0.25, P(B) = 0.35, P(C) = 0.4.$

$P(D|A) = 0.05, \quad P(D|B) = 0.04, \quad P(D|C) = 0.02.$

From the Bayes' Theorem :

$$P(A/D) = \frac{P(D/A)\times P(A)}{P(D/A)\times P(A)+P(D/B)\times P(B)+P(D/C)\times P(C)}$$

$$= \frac{0.05\times 0.25}{0.05\times 0.25+0.04\times 0.35+0.02\times 0.4}$$

$$= \frac{0.0125}{0.0125+0.014+0.008}$$

$$\mathbf{P(A/D) = 0.3621}$$

Similarly :

$$P(B/D) = \frac{P(D/B)\times P(B)}{P(D/A)\times P(A)+P(D/B)\times P(B)+P(D/C)\times P(C)}$$

$$= \frac{0.04\times 0.35}{0.05\times 0.25+0.04\times 0.35+0.02\times 0.4}$$

$$= \frac{0.014}{0.0125+0.014+0.008} = \frac{0.014}{0.0345}$$

$$\mathbf{P(B/D) = 0.4057}$$

$$P(C/D) = \frac{P(D/C)\times P(C)}{P(D/A)\times P(A)+P(D/B)\times P(B)+P(D/C)\times P(C)}$$

$$= \frac{0.02\times 0.4}{0.05\times 0.25+0.04\times 0.35+0.02\times 0.4}$$

$$= \frac{0.008}{0.0125+0.014+0.008} = \frac{0.008}{0.0345}$$

$$\mathbf{P(C/D) = 0.2318}$$

Example 3.2.2

At a certain university, 4 % of men are over 6 feet tall and 1 % of women are over 6 feet tall. The total student population is divided in the ratio 3: 2 in favour of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman?

Solution : Let us assume following :

M = {Student is Male},

F = {Student is Female},

T = {Student is over 6 feet tall}.

Given data :

$P(M) = 2/5,$

$P(F) = 3/5,$

$P(T|M) = 4/100$

$P(T|F) = 1/100.$

We require to find $P(F|T)$?

Using Bayes' Theorem we have :

$$P(F/T) = \frac{P(T/F)\,P(F)}{P(T/F)\,P(F) + P(T/M)\,P(M)} = \frac{\frac{1}{100} \times \frac{3}{5}}{\frac{1}{100} \times \frac{3}{5} + \frac{4}{100} \times \frac{2}{5}} = \frac{\frac{3}{500}}{\frac{3}{500} + \frac{8}{500}}$$

$$P(F/T) = \frac{3}{11}$$

3.2.1 Prior and Posterior Probability:

- In Bayesian learning, the best hypothesis means the most probable hypothesis, given the data D plus any initial knowledge about the prior probabilities of the various hypothesis in H.
- Bayes' theorem provides a way to calculate the probability of a hypothesis based on its prior probability, the probabilities of observing various data given the hypothesis and the observed data itself.
- Bayes' theorem is a method to revise the probability of an event given additional information.
- Bayes' theorem calculates a conditional probability called a posterior or revised probability.

- Bayes' theorem is a result in probability theory that relates conditional probabilities. If A and B denote two events, P(A|B) denotes the conditional probability of A occurring, given that B occurs. The two conditional probabilities P(A|B) and P(BA) are in general different.
- This theorem gives a relation between P(A|B) and P(BA). An important application of Bayes' theorem is that it gives a rule how to update or revise the strengths of evidence-based beliefs in light of new evidence a porteriori.
- A prior probability is an initial probability value originally obtained before any additional information is obtained.
- The prior knowledge or belief about the probabilities of various hypotheses in H is called Prior in context of Bayes' theorem.
- The probability that a particular hypothesis holds for a data set based on the Prior is called the posterior probability or simply Posterior.
- A posterior probability is a probability value that has been revised by using additional information that is later obtained.
- If A and B are two random variables

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

- In the context of classifier hypothesis h and training data I.

$$p(h/I) = \frac{P(I/h)P(h)}{P(I)}$$

Where

(h) = Prior probability of hypothesis h

(I) = Prior probability of training data I

(h|I) = Probability of h given I

P(I|h)=Probability of I given h

3.2.2 Maximum Likelihood Estimation

- Maximum - Likelihood Estimation (MLE) is a method of estimating the parameters of a statistical model. When applied to a data set and given a statistical model, maximum likelihood estimation provides estimates for the model's parameters.

$X_1, X_2, X_3, \cdots, X_n$ have joint density denoted $f_\theta(x_1, x_2, \cdots, x_n) = f(x_1, x_2, \cdots, x_n|\theta)$.
Given observed values $X_1 = X_1, x_2 = x_2, \cdots, X_n = x_n$,

$$lik(\theta) = f(x_1, x_2, \cdots, x_n|\theta)$$

Considered as a function of θ.

- If the distribution is discrete, f will be the frequency distribution function.
- The maximum likelihood estimate of θ is that value of that maximises lik(θ): It is the value that makes the observed data the most probable.

Examples of maximizing likelihood:

• A random variable with this distribution is a formalization of a coin toss. The value of the random variable is 1 with probability θ and 0 with probability **1 - θ**. Let X be a Bernoulli random variable and let x be an outcome of X, then we have

$$P(X = x) = \begin{bmatrix} \theta & \text{if } x=1 \\ 1-\theta & \text{if } x=0 \end{bmatrix}$$

Usually, we use the notation P(.) for a probability mass and the notation P(.) for a probability density. For mathematical convenience write P(X) as

$$P(X = x) = \theta^x(1-\theta)^{1-x}$$

3.3 Bayes' Theorem and Concept Learning

A consistent learner is one that returns some hypothesis h from the hypothesis class H that is consistent with a random sequence of m examples. A consistent learner is a MAP learner, if all hypothesis are a-priori equally likely.

3.3.1 Consistent Learners

- The group of learners who commit zero error over the training data and output the hypothesis are called consistent learners.
- If the training data is noise free and deterministic and if there is uniform prior probability distribution over H, then every consistent learner outputs the MAP hypothesis

3.3.2 Bayes Optimal Classifier:

- Bayes' classifier is a classifier that minimizes the error in a probabilistic manner. If it is Bayes' optimal, then the errors are weighed using the join probability distribution between the input and the output sets.

- The Bayes error is then the error of the Bayes classifier.

3.3.3 Bayes Naive Classifier:

When working with Machine Learning problems that deal with labeled training datasets, the algorithms are categorized as either classification or regression. When you go a little deeper, one of the most fundamental algorithms you will encounter is Naïve Bayes Algorithm.

Your new year resolution is to keep your health in check and so you decide to develop a running habit. Now, let's say you want to predict if you will go for a run or not based on the weather conditions outside.

So, this becomes a classification problem.

Naïve Bayes is a simple classification algorithm that can help you with your model.

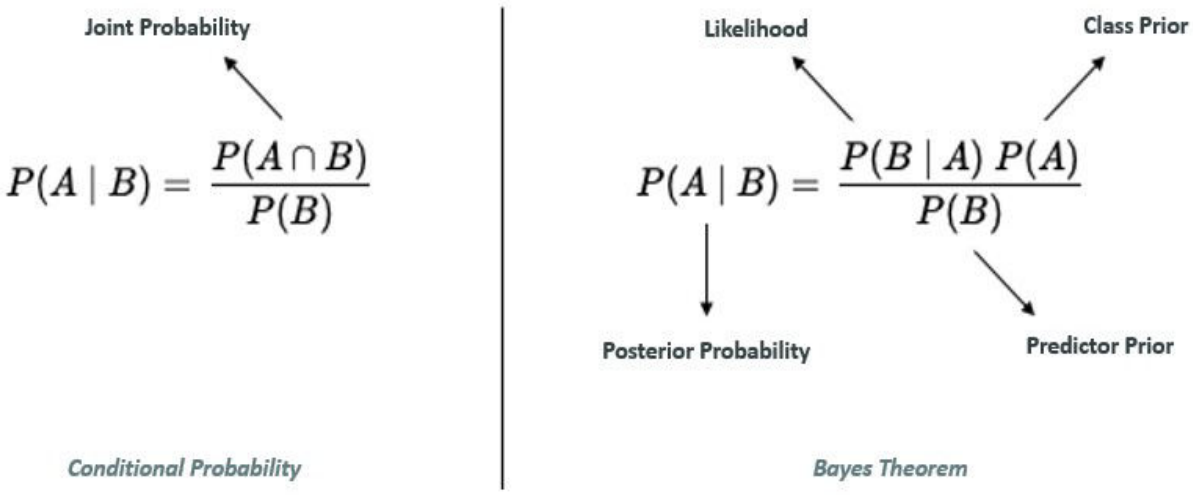
Introduction to Bayes Theorem:

The Bayes in Naïve Bayes comes from **Bayes' Theorem**.

Bayes' Theorem describes the probability of an event based on prior knowledge of the conditions that might be related to the event.

$$P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}$$

This equation is derived from the formula of conditional probability given below:



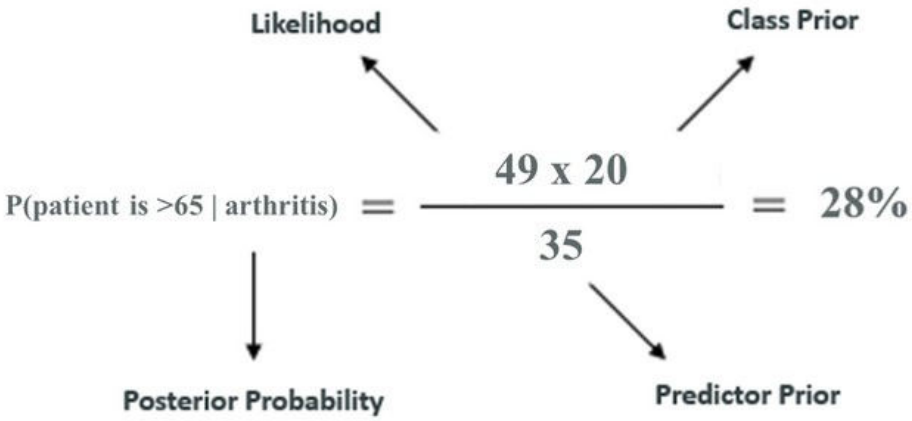
Example – Say the likelihood of a person having Arthritis if they are over 65 years of age is 49%.

Check the above stats at: Centre for Disease Control and Prevention

Now, let's assume the following:

- **Class Prior:** The probability of a person stepping in the clinic being >65-year-old is 20%
- **Predictor Prior:** The probability of a person stepping into the clinic having Arthritis is 35%

What is the probability that a person is >65 years given that he has Arthritis? This is Let's calculate this with the help of Bayes' theorem!



In fact, Bayes' Theorem is nothing but an alternate way of calculating the conditional probability of an event. When the reverse conditional probability is easier to calculate than the joint probability, Bayes' Theorem is used.

You can use Bayes' theorem to build a learner ML model, from an existing set of attributes, that predicts the probability of the response variable belonging to some class, given a new set of attributes.

Consider the previous equation again. Now, assume that event A is the response variable and event B is the input attribute. So according to the equation,

- **P(A) or Class Prior** is the prior probability of the response variable
- **P(B) or Predictor Prior** is the evidence or the probability of training data
- **P(A|B) or Posterior Probability** is the conditional probability of the response variable being of a particular value given the input attributes
- **P(B|A) or Likelihood** is basically the **reverse of the posterior probability** or the likelihood of training data

Stepwise Bayes Theorem

Let's come back to the problem at hand. Looks like you're very serious with your resolution this time given that you have been keeping track of the weather outside for the past two weeks:

Step 1 – Collect raw data

Day	Outlook	Humidity	Wind	Run
D1	Sunny	High	Weak	No
D2	Overcast	High	Strong	No
D3	Rainy	High	Weak	Yes
D4	Rainy	Normal	Weak	No
D5	Sunny	Normal	Weak	Yes
D6	Sunny	High	Weak	Yes
D7	Sunny	High	Weak	Yes
D8	Rainy	Normal	Strong	No
D9	Overcast	High	Weak	Yes
D10	Sunny	High	Weak	Yes
D11	Rainy	High	Weak	No
D12	Overcast	Normal	Strong	No
D13	Overcast	High	Weak	Yes
D14	Sunny	High	Weak	Yes

Next, you need to create a frequency table for each attribute of your dataset.

Step 2 – Convert data to a frequency table(s)

Frequency Table		Run	
		Yes	No
Outlook	Sunny	5	1
	Overcast	2	2
	Rainy	1	3

Frequency Table		Run	
		Yes	No
Humidity	High	7	3
	Normal	1	3

Frequency Table		Run	
		Yes	No
Wind	Strong	0	3
	Weak	9	2

Then, for each frequency table, you will create a likelihood table.

Step 3 – Calculate prior probability and evidence

Likelihood Table		Run		
		Yes	No	
Outlook	Sunny	5/8	1/6	6/14
	Overcast	2/8	2/6	4/14
	Rainy	1/8	3/6	4/14
		8/14	6/14	

Likelihood Table for Outlook

Likelihood Table		Run		
		Yes	No	
Humidity	High	7/8	3/6	10/14
	Normal	1/8	3/6	4/14
		8/14	6/14	

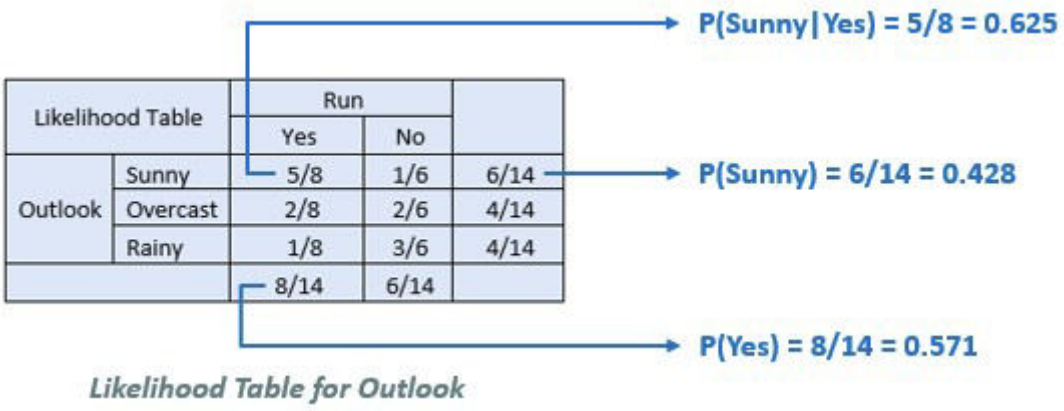
Likelihood Table for Humidity

Likelihood Table		Run		
		Yes	No	
Wind	Strong	0/9	3/5	3/14
	Weak	9/9	2/5	11/14
		9/14	5/14	

Likelihood Table for Wind

Step 4 – Apply probabilities to Bayes' Theorem equation

Let's say you want to focus on the likelihood that you go for a run given that it's sunny outside.



Likelihood Table		Run		
		Yes	No	
Outlook	Sunny	5/8	1/6	6/14
	Overcast	2/8	2/6	4/14
	Rainy	1/8	3/6	4/14
		8/14	6/14	

Likelihood Table for Outlook

P(Yes|Sunny) = (P(Sunny|Yes) * P(Yes)) / P(Sunny) = 0.625 * 0.571 / 0.428 = **0.834**

The Naive Bayes Algorithm:

Naïve Bayes assumes conditional independence over the training dataset. The classifier separates data into different classes according to the Bayes' Theorem. But assumes that the relationship between all input features in a class is independent. Hence, the model is called **naïve**.

This helps in simplifying the calculations by dropping the denominator from the formula while assuming independence:

$$\textit{Bayes Theorem} \longrightarrow P(A \mid x_1, \ldots, x_n) = \frac{P(x_1, \ldots, x_n \mid A)\, P(A)}{P(x_1, \ldots, x_n)}$$

$$\downarrow$$

$$\textit{Naïve Bayes} \longrightarrow P(A \mid x_1, \ldots, x_n) = P(x_1 \mid A) \cdot P(x_2 \mid A) \cdot P(x_i \mid A)\, P(A)$$

Let's understand this through our running resolution example:

Say you want to predict if on the coming Wednesday, given the following weather conditions, should you go for a run or sleep in:

(Outlook: Rainy,Humidity: Normal,Wind: Weak,Run: ?)

Likelihood of 'Yes' on Wednesday:

P(Outlook = Rainy|Yes) * P(Humidity = Normal|Yes) * P(Wind = Weak|Yes) * P(Yes)

= 1/8 * 1/8 * 9/9 * 8/14 = **0.0089**

Likelihood of 'No' on Wednesday:

P(Outlook = Rainy|No) * P(Humidity = Normal|No) * P(Wind = Weak|No) * P(No)

= 3/6 * 3/6 * 2/5 * 6/14 = **0.0428**

Now, to determine the probability of going for a run on Wednesday, you just need to divide P(Yes) with the sum of the likelihoods of Yes and No.

P(Yes) = 0.0089 / (0.0089 + 0.0428) = 0.172

Similarly, P(No) = 0.0428 / (0.0089 + 0.0428) = 0.827

According to your model, it looks like there's an almost 83% probability that **you're going to stay under the covers next Wednesday!**

Although Naïve Bayes is used for weather predictions, for advanced machine learning problems, the complexity of the Bayesian classifier needs to be reduced for it to be practical. This is where the naïve in Naïve Bayes comes in.

Industrial Applications of Naïve Bayes

- **Text Classification:** Naïve Bayes algorithm is almost always used as a classifier and is an excellent choice for spam filtering of your emails or news categorization on your smartphone.
- **Recommendation Systems:** Naïve Bayes is used with collaborative filtering to build recommendation systems for you. The '*Because you watched* ____' section on Netflix is exactly that.
- **Sentiment Analysis:** Naïve Bayes is an effective algorithm for the identification of positive or negative sentiments of a target group (customers, audience, etc.). Think of feedback forms and IMDb reviews.

Types of Naïve Bayes Classifiers

Bernoulli Naïve Bayes

- Predictors are Boolean variables
- Used when data is as per multivariate Bernoulli distribution
- Popular for discrete features

Multinomial Naïve Bayes

- Uses frequency of present words as features
- Commonly used for document classification problems
- Popular for discrete features as well

Gaussian Naïve Bayes

- Used when data is as per the Gaussian distribution
- Predictors are continuous variables

Advantages of Naïve Bayes

- Easy to work with when using binary or categorical input values.
- Require a small number of training data for estimating the parameters necessary for classification.
- Handles both continuous and discrete data.
- Fast and reliable for making real-time predictions.

Limitations of Naive Bayes

- Assumes that all the features are independent, which is highly unlikely in practical scenarios.
- Unsuitable for numerical data.
- The number of features must be equal to the number of attributes in the data for the algorithm to make correct predictions.
- Encounters 'Zero Frequency' problem: If a categorical variable has a category in the test dataset that wasn't included in the training dataset, the model will assign it a 0

probability and will be unable to make a prediction. This problem can be resolved using smoothing techniques which are out of the scope of this article.

- Computationally expensive when used to classify a large number of items.

31/08/23

Naive Bayes classifier.

* Consider following dataset and predict the class of new instance using Naive Bayes classification algorithm.

Day	Outlook	Temperature	Humidity	wind	Play Tennis
D_1	Sunny	Hot	High	Weak	NO
D_2	Sunny	Hot	High	Strong	NO
D_3	Overcast	Hot	High	Weak	Yes
D_4	Rain	mild	High	Weak	Yes
D_5	Rain	cool	Normal	Weak	yes
D_6	Rain	cool	Normal	strong	NO
D_7	Overcast	cool	Normal	Strong	Yes
D_8	Sunny	mild	High	weak	No
D_9	Sunny	cool	Normal	Weak	Yes
D_{10}	Rain	mild	Normal	Weak	Yes
D_{11}	Sunny	mild	Normal	Strong	Yes
D_{12}	Overcast	mild	High	Strong	Yes
D_{13}	Overcast	Hot	Normal	Weak	Yes
D_{14}	Rain	mild	High	Strong	NO

new instance = (outlook = Sunny,
temperature = cool
Humidity = High
wind = strong)

Sol:- The aim of Naive Bayes classifier to classify the Example in one of the given categories.

Step1 :- calculate prior probability,

$P(\text{Play Tennis} = \text{Yes}) = \frac{9}{14}$

$P(\text{play Tennis} = \text{No}) = \frac{5}{14}$

step2 :-

calculate conditional probabilities of individual attributes outlook, temperature, Humidity, wind.

Outlook	Yes	NO
Sunny	$\frac{2}{9}$	$\frac{3}{5}$
Overcast	$\frac{4}{9}$	0
Rain	$\frac{3}{9}$	$\frac{2}{5}$

Temperature	Yes	NO
Hot	$\frac{2}{9}$	$\frac{2}{5}$
mild	$\frac{4}{9}$	$\frac{2}{5}$
cool	$\frac{3}{9}$	$\frac{1}{5}$

Humidity	Yes	NO
High	$\frac{3}{9}$	$\frac{4}{5}$
Normal	$\frac{6}{9}$	$\frac{1}{5}$

wind	Yes	NO
Strong	$\frac{3}{9}$	$\frac{3}{5}$
weak	$\frac{6}{9}$	$\frac{2}{5}$

step3 :-

Given the new instance $\vec{x}$ = (outlook = sunny, Temperature = cool, Humidity = High, wind = strong)

The naive Bayes classifier Equation is;

$$V_{NB} = \underset{v_j \in [Yes, NO]}{\operatorname{argmax}} \; P(v_j) \prod P(a_i | v_j)$$

Prior probability · Product

v_j → Target Variable.

$P(a_i | v_j)$ → posterior probability

$V_{NB}(Yes) = P(Yes) \cdot P(sunny|yes) \cdot P(cool|yes) \cdot P(High|yes)$
$P(Strong|Yes)$

$= \frac{9}{14} \times \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9}$

$= 0.0053$

$V_{NB}(NO) = P(NO) \cdot P(sunny/NO) \cdot P(cool|NO) \cdot P(High/NO) \cdot P(strong|NO)$

$= \frac{5}{14} \times \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5}$

$= 0.0206$

Step 4 :- we need to normalise the probabilities.

The normalized values can be calculated like this

$$V_{NB}(Yes) = \frac{V_{NB}(Yes)}{V_{NB}(Yes) + V_{NB}(NO)} \Rightarrow \frac{0.0053}{0.0053 + 0.0206}$$

$= 0.205$

$$V_{NB}(NO) = \frac{V_{NB}(NO)}{V_{NB}(Yes) + V_{NB}(NO)} \Rightarrow \frac{0.0206}{0.0053 + 0.0206}$$

$= 0.795$

$\Rightarrow$ Since $P(NO) > P(Yes)$, Hence the new instance is classified as "NO". Here the meaning is the person will not play Tennis.

Ex2:-

1. Estimate conditional probabilities of each attributes {color, Legs, height, smelly} for the species classes: {M, H} using the data given in the Table.

2. Using these probabilities estimate the probability values for the new instance (color = green, legs = 2, Height = Tall, Smelly = No)

NO	color	Legs	Height	Smelly	species
1	White	3	Short	Yes	M
2	Green	2	Tall	NO	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	NO	H
6	White	2	Tall	NO	H
7	White	2	Tall	NO	H
8	White	2	Short	Yes	H

sol:- step1 :-

Prior probabilities

$P(M) = \frac{4}{8}$

$P(H) = \frac{4}{8}$

step2 :-

conditional probabilities of attributes;

color	M	H
White	$\frac{2}{4}$	$\frac{3}{4}$
Green	$\frac{2}{4}$	$\frac{1}{4}$

Legs	M	H
3	$\frac{3}{4}$	0
2	$1/4$	$\frac{4}{4}-1$

Height	M	H
Short	$\frac{3}{4}$	$\frac{2}{4}$
Tall	$\frac{1}{4}$	$\frac{2}{4}$

Smelly	M	H
Yes	$\frac{3}{4}$	$\frac{1}{4}$
NO	$\frac{1}{4}$	$\frac{3}{4}$

Step3 :-

Given new instance is;

(color = green, legs = 2, Height = Tall, smelly = No)

$$P(M/\text{new instance}) = P(M) * P(\text{color}=\text{green}/M) * P(\text{legs}=2/M) * P(\text{Height}=\text{Tall}/M) * P(\text{smelly}=\text{No}/M)$$

$$= \frac{4}{8} * \frac{2}{4} * \frac{1}{4} * \frac{1}{4} * \frac{1}{4}$$

$$= 0.003906$$

$$P(H/\text{new instance}) = P(H) * P(\text{color}=\text{green}/H) * P(\text{legs}=2/H) * P(\text{Height}=\text{Tall}/H) * P(\text{smelly}=\text{No}/H)$$

$$= \frac{4}{8} \times \frac{1}{4} \times 1 \times \frac{2}{4} \times \frac{3}{4}$$

$$= 0.047$$

Since $P(H/\text{new instance}) > P(M/\text{new instance})$, Hence the new instance belongs to species H.

Ex-3:-

Example NO.	color	Type	Origin	stolen?
1	Red	sports	Domestic	Yes
2	Red	sports	Domestic	NO
3	Red	sports	Domestic	Yes
4	Yellow	Sports	Domestic	NO
5	Yellow	sports	Imported	Yes
6	Yellow	SUV	Imported	NO
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	NO
9	Red	SUV	Imported	NO
10	Red	sports	Imported	Yes

Q): New instance = (Red, SUV, Domestic)

Sol: step1:

Prior probability

$P(Yes) = \frac{5}{10} = \frac{1}{2}$; $P(NO) = \frac{5}{10} = \frac{1}{2}$.

step2:

conditional probabilities

Color	Yes	NO
Red	$\frac{3}{5}$	$\frac{2}{5}$
Yellow	$\frac{2}{5}$	$\frac{3}{5}$

Type	Yes	NO
Sports	$\frac{4}{5}$	$\frac{2}{5}$
SUV	$\frac{1}{5}$	$\frac{3}{5}$

Origin	Yes	NO
Domestic	$\frac{2}{5}$	$\frac{3}{5}$
Imported	$\frac{3}{5}$	$\frac{2}{5}$

Step3:- Given new instance is; (Red, SUV, Domestic)

$P(Yes/new\ instance) = P(Yes) * P(Color=Red/Yes) * P(SUV Type=SUV/Yes) * P(Origin=Domestic/Yes)$

$= \frac{1}{2} \times \frac{3}{5} \times \frac{1}{5} \times \frac{2}{5}$

$= 0.024$

$P(NO/new\ instance) = P(NO) * P(color=Red/NO) * P(Type=SUV/Yes) * P(origin=Domestic/yes)$

$= \frac{1}{2} \times \frac{2}{5} \times \frac{3}{5} \times \frac{3}{5}$

$= 0.072$

since $P(NO/new\ instance) > P(Yes/new\ instance)$, Hence we can classify this model as "NO" in this case.

3.4 Bayesian Belief Network

- Bayesian Belief Networks (BBN) are also known as belief networks, Bayesian networks, and probabilistic networks. BBN is a special type of diagram (called a directed graph) together with an associated set of probability tables.
- The graph consists of nodes and arcs. The nodes represent variables, which can be discrete or continuous. The arcs represent causal relationships between variables.
- A belief network is defined by two components: Directed acyclic graph and a set of conditional probability tables.
- BBNs enable us to model and reason about uncertainty. BBNs accommodate both subjective probabilities and probabilities based on objective data. The most important use of BBNs is in revising probabilities in the light of actual observations of events.
- Each node in the directed acyclic graph represents a random variable. The variables may be discrete or continuous valued. They may correspond to actual attributes given in the data or to "hidden variables" believed to form a relationship.
- Each arc represents a probabilistic dependence. If an arc is drawn from a node Y to a node Z, then Y is a parent or immediate predecessor of Z, and Z is a descendant of Y. Each variable is conditionally independent of its non descendants in the graph, given its parents.
- The below Figure shows simple belief network.

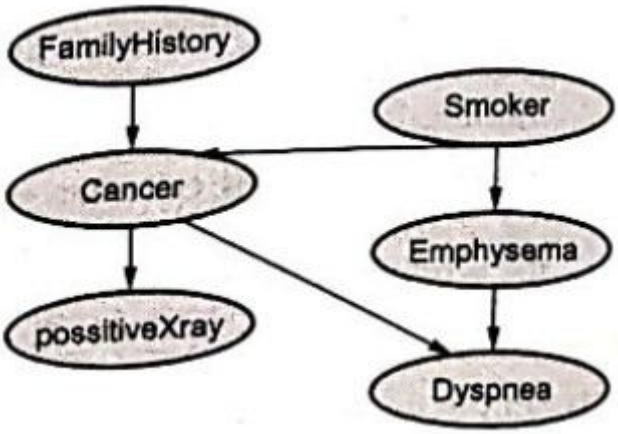


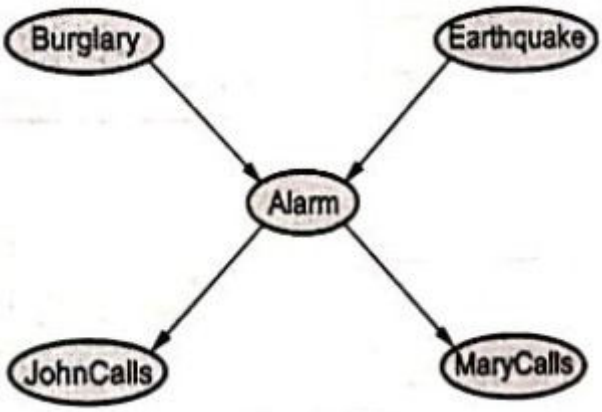
Figure: Simple belief network

- The diagram consists of nodes and arcs. The nodes represent the discrete continuous variables for which we interested to calculate the conditional probabilities. The arc represents the causal relationship of the variables.
- Belief network has one Conditional Probability Table (CPT) for each variable.
- Probability theory is the body of knowledge that enables us to reason formally about uncertain events.

- The probability P of an uncertain event A, written P(A) is defined by the frequency of that event based on previous observations. This is called frequency based probability.
- In general, a person belief in a statement a will depend on some body of knowledge K. We write this as P(a|K). The expression P(a|K) thus represents a belief measure.
- Sometimes, for simplicity, when K remains constant we just write P(a), but you must be aware that this is a simplification.
- The notion of degree of belief P(A|K) is an uncertain event A is conditional on a body of knowledge K. In general, we write P(A|B) to rep represent a belief in A under the assumption that B is known.
- Bayesian belief network describes the joint probability distribution of a set of attributes in their joint space.
- Bayesian networks are used for modelling beliefs in domains like computational biology and bioinformatics such as protein structure and gene regulatory networks, medicines, forensics, document classification, information retrieval, image processing, decision support systems, sports betting and gaming.

Example: Alarm system example.

- Assume your house has an alarm system against burglary. You live in the seismically active area and the alarm system can get occasionally set off by an earthquake.
- You have two neighbors, Mary and John, who do not know each other. If they hear the alarm they call you, but this is not guaranteed.
- We want to represent the probability distribution of events: Burglary, Earthquake, Alarm, Mary calls and John calls.
- **Causal relations :**



Directed acyclic graph:

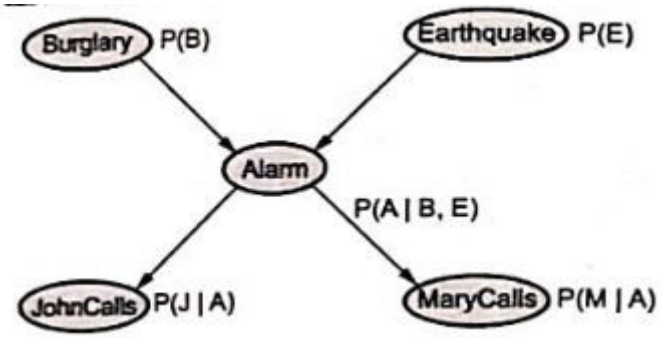
- Nodes = Random variables

Burglary, Earthquake, Alarm, Mary calls and John calls

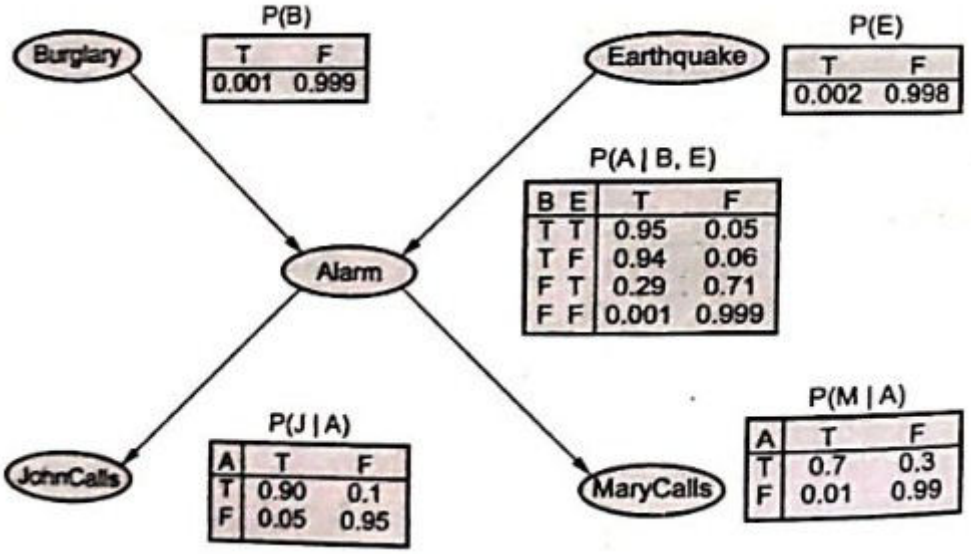
- Links = Direct (causal) dependencies between variables

  The chance of Alarm is influenced by Earthquake, The chance of John calling is affected by the Alarm.

  Local conditional distributions: Relate variables and their parents.



Bayesian belief network:



PYTHON EXERCISE ON NAIVE BAYES

1. Assuming a set of documents that need to be classified, use the naïve Bayesian Classifier model to perform this task. write the python program for calculating the accuracy, precision and recall for your data set.

Naive Bayes algorithms for learning and classifying text

LEARN_NAIVE_BAYES_TEXT (Examples, V)

Examples is a set of text documents along with their target values. V is the set of all possible target values. This function learns the probability terms $P(w_k|v_j,)$, describing the probability that a randomly drawn word from a document in class v_j will be the English word w_k. It also learns the class prior probabilities $P(v_j)$.

1. *collect all words, punctuation, and other tokens that occur in Examples*
   - *Vocabulary* ← ***c*** the set of all distinct words and other tokens occurring in any text document from *Examples*

2. *calculate the required $P(v_j)$ and $P(w_k|v_j)$ probability terms*

   - For each target value v_j in *V* do
     - $docs_j$ ← the subset of documents from *Examples* for which the target value is *vj*
     - $P(v_j)$ ← $|docs_j| / |Examples|$
     - $Text_j$ ← a single document created by concatenating all members of $docs_j$
     - n ← total number of distinct word positions in $Text_j$
     - for each word w_k in *Vocabulary*
       - n_k ← number of times word $\boldsymbol{w_k}$ occurs in $Text_j$
       - $P(w_k|v_j)$ ← $(n_k + 1) / (n + |Vocabulary|)$

CLASSIFY_NAIVE_BAYES_TEXT (Doc)

Return the estimated target value for the document Doc. a_i denotes the word found in the ith position within Doc.

- *positions* ← all word positions in *Doc* that contain tokens found in *Vocabulary*
- Return V_{NB}, where

$$v_{NB} = \underset{v_j \in V}{\operatorname{argmax}} P(v_j) \prod_{i \in positions} P(a_i|v_j)$$

Data set:

	Text Documents	Label
1	I love this sandwich	pos
2	This is an amazing place	pos
3	I feel very good about these beers	pos
4	This is my best work	pos
5	What an awesome view	pos
6	I do not like this restaurant	neg
7	I am tired of this stuff	neg
8	I can't deal with this	neg
9	He is my sworn enemy	neg
10	My boss is horrible	neg
11	This is an awesome place	pos
12	I do not like the taste of this juice	neg
13	I love to dance	pos
14	I am sick and tired of this place	neg
15	What a great holiday	pos
16	That is a bad locality to stay	neg
17	We will have good fun tomorrow	pos
18	I went to my enemy's house today	neg

Program:

```
import pandas as pd

msg=pd.read_csv('naivetext.csv',names=['message','label'])

print('The dimensions of the dataset',msg.shape)

msg['labelnum']=msg.label.map({'pos':1,'neg':0})
X=msg.message
y=msg.labelnum

print(X)
print(y)

#splitting the dataset into train and test data
from sklearn.model_selection import train_test_split
xtrain,xtest,ytrain,ytest=train_test_split(X,y)

print ('\n The total number of Training Data :',ytrain.shape)
print ('\n The total number of Test Data :',ytest.shape)


#output of count vectoriser is a sparse matrix
from sklearn.feature_extraction.text import CountVectorizer
count_vect = CountVectorizer()
xtrain_dtm = count_vect.fit_transform(xtrain)
xtest_dtm=count_vect.transform(xtest)
print('\n The words or Tokens in the text documents \n')
print(count_vect.get_feature_names())

df=pd.DataFrame(xtrain_dtm.toarray(),columns=count_vect.get_fe
ature_names())

# Training Naive Bayes (NB) classifier on training data.
from sklearn.naive_bayes import MultinomialNB
clf = MultinomialNB().fit(xtrain_dtm,ytrain)
predicted = clf.predict(xtest_dtm)

#printing accuracy, Confusion matrix, Precision and Recall
from sklearn import metrics
print('\n Accuracy of the classifer is',
metrics.accuracy_score(ytest,predicted))
```

```
print('\n Confusion matrix')
print(metrics.confusion_matrix(ytest,predicted))

print('\n The value of Precision' ,
metrics.precision_score(ytest,predicted))

print('\n The value of Recall' ,
metrics.recall_score(ytest,predicted))
```

Output:

The dimensions of the dataset (18, 2)

0	I love this sandwich
1	This is an amazing place
2	I feel very good about these beers
3	This is my best work
4	What an awesome view
5	I do not like this restaurant
6	I am tired of this stuff
7	I can't deal with this
8	He is my sworn enemy
9	My boss is horrible
10	This is an awesome place
11	I do not like the taste of this juice
12	I love to dance
13	I am sick and tired of this place
14	What a great holiday
15	That is a bad locality to stay
16	We will have good fun tomorrow
17	I went to my enemy's house today

Name: message, dtype: object
0 1
1 1
2 1
3 1
4 1
5 0
6 0
7 0
8 0
9 0
10 1
11 0
12 1
13 0
14 1
15 0
16 1
17 0
Name: labelnum, dtype: int64

The total number of Training Data: (13,)

The total number of Test Data: (5,)

The words or Tokens in the text documents

['about', 'am', 'amazing', 'an', 'and', 'awesome', 'beers', 'best', 'can', 'deal', 'do', 'enemy', 'feel', 'fun', 'good', 'great', 'have', 'he', 'holiday', 'house', 'is', 'like', 'love', 'my', 'not', 'of', 'place', 'restaurant', 'sandwich', 'sick', 'sworn', 'these', 'this', 'tired', 'to', 'today', 'tomorrow', 'very', 'view', 'we', 'went', 'what', 'will', 'with', 'work']

Accuracy of the classifier is 0.8

Confusion matrix

[[2 1]

[0 2]]

The value of Precision 0.6666666666666666

The value of Recall 1.0



Basic knowledge

Confusion Matrix

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

True positives: data points labelled as positive that are actually positive

False positives: data points labelled as positive that are actually negative

True negatives: data points labelled as negative that are actually negative

False negatives: data points labelled as negative that are actually positive

$$\text{Recall} = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

$$= \frac{True\ Positive}{Total\ Actual\ Positive}$$

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

$$\text{Precision} = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

$$= \frac{True\ Positive}{Total\ Predicted\ Positive}$$

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Example:

		Actual	
		Positive	Negative
Predicted	Positive	1 TP	3 FP
	Negative	0 FN	1 TN

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{1}{1+3} = 0.25$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{1}{1+0} = 1$$

Accuracy: how often is the classifier correct?

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total}} = \frac{1+1}{5} = 0.4$$

Example: Movie Review

Doc	Text	Class
1	I loved the movie	+
2	I hated the movie	-
3	a great movie. good movie	+
4	poor acting	-
5	great acting. good movie	+

Unique word

< I, loved, the, movie, hated, a, great, good, poor, acting>

Doc	I	loved	the	movie	hated	a	great	good	poor	acting	Class
1	1	1	1	1							+
2	1		1	1	1						-
3				2		1	1	1			+
4									1	1	-
5				1			1	1		1	+

Doc	I	loved	the	movie	hated	a	great	good	poor	acting	Class
1	1	1	1	1							+
3				2		1	1	1			+
5				1			1	1		1	+

$$P(+) = \frac{3}{5} = 0.6$$

$$P(I\,|+) = \frac{1+1}{14+10} = 0.0833$$

$$P(a\,|+) = \frac{1+1}{14+10} = 0.0833$$

$$P(loved\,|+) = \frac{1+1}{14+10} = 0.0833$$

$$P(great\,|+) = \frac{2+1}{14+10} = 0.125$$

$$P(the\,|+) = \frac{1+1}{14+10} = 0.0833$$

$$P(good\,|+) = \frac{2+1}{14+10} = 0.125$$

$$P(movie\,|+) = \frac{4+1}{14+10} = 0.2083$$

$$P(poor\,|+) = \frac{0+1}{14+10} = 0.0416$$

$$P(hated\,|+) = \frac{0+1}{14+10} = 0.0416$$

$$P(acting\,|+) = \frac{1+1}{14+10} = 0.0833$$

Doc	I	loved	the	movie	hated	a	great	good	poor	acting	Class
2	1		1	1	1						-
4									1	1	-

$$P(-) = \frac{2}{5} = 0.4$$

$$P(I\,|-) = \frac{1+1}{6+10} = 0.125$$

$$P(a\,|-) = \frac{0+1}{6+10} = 0.0625$$

$$P(loved\,|-) = \frac{0+1}{6+10} = 0.0625$$

$$P(great\,|-) = \frac{0+1}{6+10} = 0.0625$$

$$P(the\,|-) = \frac{1+1}{6+10} = 0.125$$

$$P(good\,|-) = \frac{0+1}{6+10} = 0.0625$$

$$P(movie|-) = \frac{1+1}{6+10} = 0.125$$

$$P(poor|-) = \frac{1+1}{6+10} = 0.125$$

$$P(hated\,|-) = \frac{1+1}{6+10} = 0.125$$

$$P(acting|-) = \frac{1+1}{6+10} = 0.125$$

Let's classify the new document

I hated the poor acting

If $V_j = +$

then,

$= P(+)\ P(I \mid +)\ P(hated \mid +)\ P(the \mid +)\ P(poor \mid +)\ P(acting \mid +)$

$= 0.6 * 0.0833 * 0.0416 * 0.0833 * 0.0416 * 0.0833$

$= 6.03\ X\ 10^{-2}$

If $V_j = -$

then,

$= P(-)\ P(I \mid -)\ P(hated \mid -)\ P(the \mid -)\ P(poor \mid -)\ P(acting \mid -)$

$= 0.4 * 0.125 * 0.125 * 0.125 * 0.125 * 0.125$

$= 1.22\ X\ 10^{-5}$

$= 1.22\ X\ 10^{-5} > 6.03\ X\ 10^{-2}$

So, the new document belongs to $(-)$ class